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Natural Resource Stewardship and Science

Everglades National Park Recreational Saltwater Fisheries Report 2019

Natural Resource Data Series NPS/EVER/NRDS—2021/1329

ON THE COVER
Tarpon caught in Everglades National Park.
Photo courtesy of Gavin McKenzie

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Natural Resource Data Series NPS/EVER/NRDS—2021/1329

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The National Park Service, Natural Resource Stewardship and Science office in Fort Collins, Colorado, publishes a range of reports that address natural resource topics. These reports are of interest and applicability to a broad audience in the National Park Service and others in natural resource management, including scientists, conservation and environmental constituencies, and the public.

The Natural Resource Data Series is intended for the timely release of basic data sets and data summaries. Care has been taken to assure accuracy of raw data values, but a thorough analysis and interpretation of the data has not been completed. Consequently, the initial analyses of data in this report are provisional and subject to change.

All manuscripts in the series receive the appropriate level of peer review to ensure that the information is scientifically credible, technically accurate, appropriately written for the intended audience, and designed and published in a professional manner.

Data in this report were collected and analyzed using methods based on established, peer-reviewed protocols and were analyzed and interpreted within the guidelines of the protocols.

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Executive Summary

Sport fisheries in Everglades National Park showed remarkable increases in catch per unit effort (CPUE) of preferred species in 2019. Total reported catch increased 19% from 2018 to 2019, while the number of interviewed fishing trips decreased by 4%. The lower number of interviews can be explained by missing data from January 2019 due to a federal government shutdown. Average anglers per fishing trip also decreased at the same time; therefore, total fishing effort reported was lower.

This report provides an analysis of five species of fish of recreational interest: Common Snook, Red Drum, Spotted Seatrout, Gray Snapper, and Tarpon. These species represent the preferred fish catch, either stated or preferentially kept, according to angler interviews, and comprised nearly 65% of all fish reported caught recreationally in 2019. Additionally, the status of several other species or families of fish are described.

The Common Snook, *Centropomus undecimalis*, was the most frequently caught fish in both years, representing 21% of all reported catch in 2018 and increasing to 29% of total reported catch in 2019. The increase in Snook catch in 2018 and 2019 marks a significant recovery since the mass mortality of the species in South Florida coastal waters that occurred with the extreme low temperatures of January 2010. CPUE increased to the highest ever recorded for the species over the 60-year period of record. Anecdotally, most of the Snook caught were described as “under slot” (smaller than legal limit) individuals. Average Snook catch nearly reached one fish per hour on trips when it was caught, which is three times higher than the long-term average.

The other four species of fish studied in detail either remained stable or improved. Red Drum (*Sciaenops ocellata*) showed a large increase in CPUE in 2019, double the average CPUE of the previous 5 years. The catch rate for Red Drum increased 36% over 2018 levels. Spotted Seatrout (*Cynoscion nebulosus*) CPUE did not change significantly from 2018 to 2019. Gray Snapper (*Lutjanus griseus*) CPUE in 2019 remained similar to 2018. Gray Snapper (also known as Mangrove Snapper) catch rate had experienced a decadal trend upward and reached a 35-year high in 2017. Tarpon (*Megalops atlantica*), a notoriously difficult fish to catch, increased in CPUE from 2018 to 2019, continuing an eight-year trend of increasing CPUE from a low point in 2011. Tarpon are intolerant of cold temperatures, and the population suffered losses during the January 2010 cold weather in South Florida.

This report also summarizes information about other fish caught in the park’s estuarine waters. Sixty-seven total fish species were identified and reported caught by hook and line fishing during the year in 2019, but only 21 species represented 99% of the total catch. The vast majority of “other-fish-caught,” and generally not preferred, were catfish (Hardheads and Gafftopsail), Ladyfish and Crevalle Jack.

Of note in 2019, Goliath Grouper, *Epinephelus itajara*, was reported caught 964 times, an increase of 223% from 2018 and an increase of more than 1300% over the average from 2010 to 2017 (approximately 70 annually). The majority of Goliath Grouper were caught from boats fishing in Gulf of Mexico coastal waters. Harvest and possession of Goliath Grouper has been prohibited in State and Federal waters of Florida since 1990 due to near extirpation of the Florida population from

overfishing. Goliath Grouper were also negatively impacted by exceptionally cold weather in south Florida in 2010, and the substantial increases in 2019 catch represent a positive indication for the South Florida, Gulf of Mexico population.

For the elasmobranch species in the park (sharks, rays and related fish), four hundred fifty-four sharks were reported caught during recreational fishing in Everglades National Park in 2019, down 36% from 2018. Most of the sharks caught were Blacktip, Bonnethead, Bull and Lemon Sharks. Only three sharks were harvested by anglers. Lemon sharks, *Negaprion brevirostris*, along with 26 other shark species are "prohibited-take" in Florida waters. Eleven Smalltooth Sawfish, *Pristis pectinata*, were reported caught by hook and line incidentally in 2019. The Smalltooth Sawfish is endangered and protected by the Endangered Species Act. Everglades National Park coastal waters are designated as critical habitat for the species in the United States.

Florida waters have experienced a rapid expansion of invasive Lionfish (*Pterois volitans*) over the past decade. Densities of the introduced species from the Pacific Ocean on Atlantic reefs and along coastal Gulf of Mexico waters have increased to alarming levels. However, preliminary surveys of Florida Bay indicate the Lionfish has not established in Everglades National Park marine waters to a substantial degree. A total of ten juvenile Lionfish were removed from the park in 2019, a reduction from the 35 taken the preceding year.

The interview data included in this report represent angler activity collected from Flamingo and Everglades City. Approximately equal numbers of reports come from the two Creel interview sites. A total of 2461 trips were interviewed in 2019. The average number of anglers per trip has been decreasing for the past five years and reached 2.17 in 2019. Anglers from the local area normally outnumber those from other parts of Florida. The proportions switched in 2019, with anglers categorized as outside-the-local-area reported as the largest category of interviews (51%) for the first time in the period of record.

Acknowledgments

Grateful acknowledgement is extended to Tylan Dean and Antonia Florio for review and comments on the draft versions of this document. Thanks also to Gage Beckley, Brian Hutchinson, and Grace Hevey who assisted with collection and computer entry of angler interview data.

Introduction and Methods

Recreational fishing in Everglades National Park coastal waters has been a premier interest of Florida residents and out-of-state visitors, and it has supported a substantial segment of the South Florida economy for decades. Marine and estuarine waters of Everglades National Park fall within the Marjory Stoneman Douglas Wilderness. The submerged land, emergent islands, and coastal lands are all designated wilderness with special protection, but the marine and tidally influenced water itself is excluded from this designation. Thus, motorized conveyance and saltwater fishing is allowed in Everglades National Park waters and estuarine tributaries in the Gulf of Mexico, Florida Bay and Whitewater Bay. Preferred catch in Everglades coastal waters consists of several game fish including Common Snook, Red Drum, Spotted Seatrout, Gray Snapper, and Tarpon. These species are all naturally present within park boundaries.

Park staff indirectly monitor sport fish population trends through regular interviews of recreational anglers by asking them: 1) how long they fished, 2) their preferred species, 3) which species and how many were caught and kept, and 4) where they fished. Staff attempt to sample 30% of the anglers fishing marine and tidal areas within park boundaries on a given weekend. This information is collected at Flamingo marina, for Florida Bay and Whitewater Bay, and at Everglades City and Chokoloskee marinas on the Gulf of Mexico. The interviews, historically referred to as Creel surveys, have been collected in their current format weekly since 1972. A similar, but more limited, survey was performed from 1958 to 1967, and a period of no interviews occurred between 1968 to 1971. Other than periodic interruptions with duration of several consecutive weeks to months (due to hurricanes or other external interventions), surveys have been performed in the same manner on each weekend continuously at each of these sites since 1980. Data reported in this document do not include catch by anglers fishing within Everglades National Park through commercial vendors (fishing guides), and cannot account for anglers that embark from, and return to, docks in the Florida Keys.

In this report, catch is normalized as catch per unit effort, or CPUE, which is defined as the number of fish of a species caught by hook and line per fishing hour of all individuals fishing from a boat within Everglades National Park boundaries, as reported by anglers at the end of a fishing trip. The park marine and estuarine waters are sectioned into three ecologically distinct areas, namely Florida Bay and Cape Sable (FBCS), Whitewater Bay (WWB) and Gulf Coast (GC), identified in Figure 1. Catch is categorized by area based upon the primary location fished during the day.

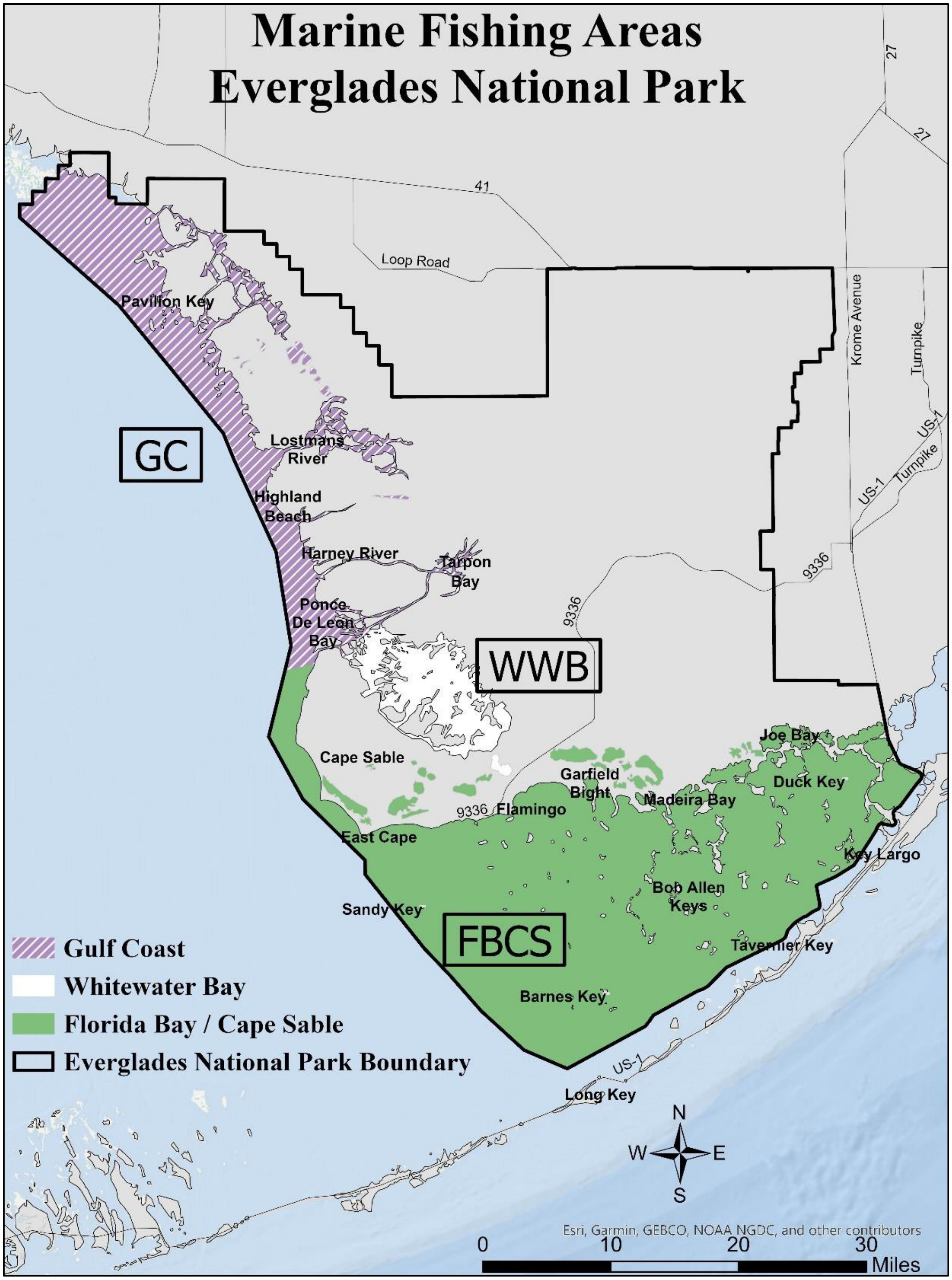

Figure 1. Map of the boundaries of Everglades National Park (bold black line) three ecologically distinct fishing areas represent Florida Bay and Cape Sable (FBCS), Whitewater Bay (WWB), and Gulf Coast (GC) areas.

Results and Discussion

Overall Catch and Preferred Species CPUE

Thirty-two families, and at least sixty-seven species of finfish were caught and reported for Everglades coastal marine and estuarine waters in 2019, including various catfish, jacks, sharks and rays, snapper, seatrout, drum, mackerel and grouper. The catch sampled from Everglades National Park recreational fishing in 2019 totaled 61,840 fish, an increase of 19% over 2018 total catch reported (51,960). Interview effort was similar between the two years.

The family Centropomidae, represented by a single species, the Common Snook (referred to simply as Snook), made up 29% of the total catch amounting to 17,869 fish. The family Sciaenidae including seatrout and drum accounted for another 27% totaling 16,900 fish. The families Ariidae (Hardhead and Gafftopsail Catfish) – 7,351 fish, Elopidae (Ladyfish and Atlantic Tarpon) – 6,133 fish, Carangidae (e.g., jacks and pompano) – 5,191 fish, and Lutjanidae (snappers) – 5,033 fish, each made up greater than 8% of the annual catch. The six families (Centropomidae, Sciaenidae, Ariidae, Elopidae, Carangidae, and Lutjanidae) comprised nearly 95% of all fish caught in 2019 (see Appendix 1). Serranidae (groupers) totaled 988 fish or 1.6% of the total catch.

Five preferred recreational saltwater fish species comprised 39,998 fish or 65% of the total reported catch in 2019. Preferred catch refers to species of stated preference (Snook, Red Drum and Tarpon) or those preferentially kept (Spotted Seatrout and Gray Snapper). The number of fish of a given species caught per angler hour, or CPUE, of preferred fish varied by species, and either increased or had no substantial change from 2018 to 2019 (Figure 2). CPUE increased significantly for Snook (42%) and Red Drum (36%) and Tarpon (30%) from 2018 to 2019. Spotted Seatrout, Gray Snapper CPUE remained relatively stable over the last year.

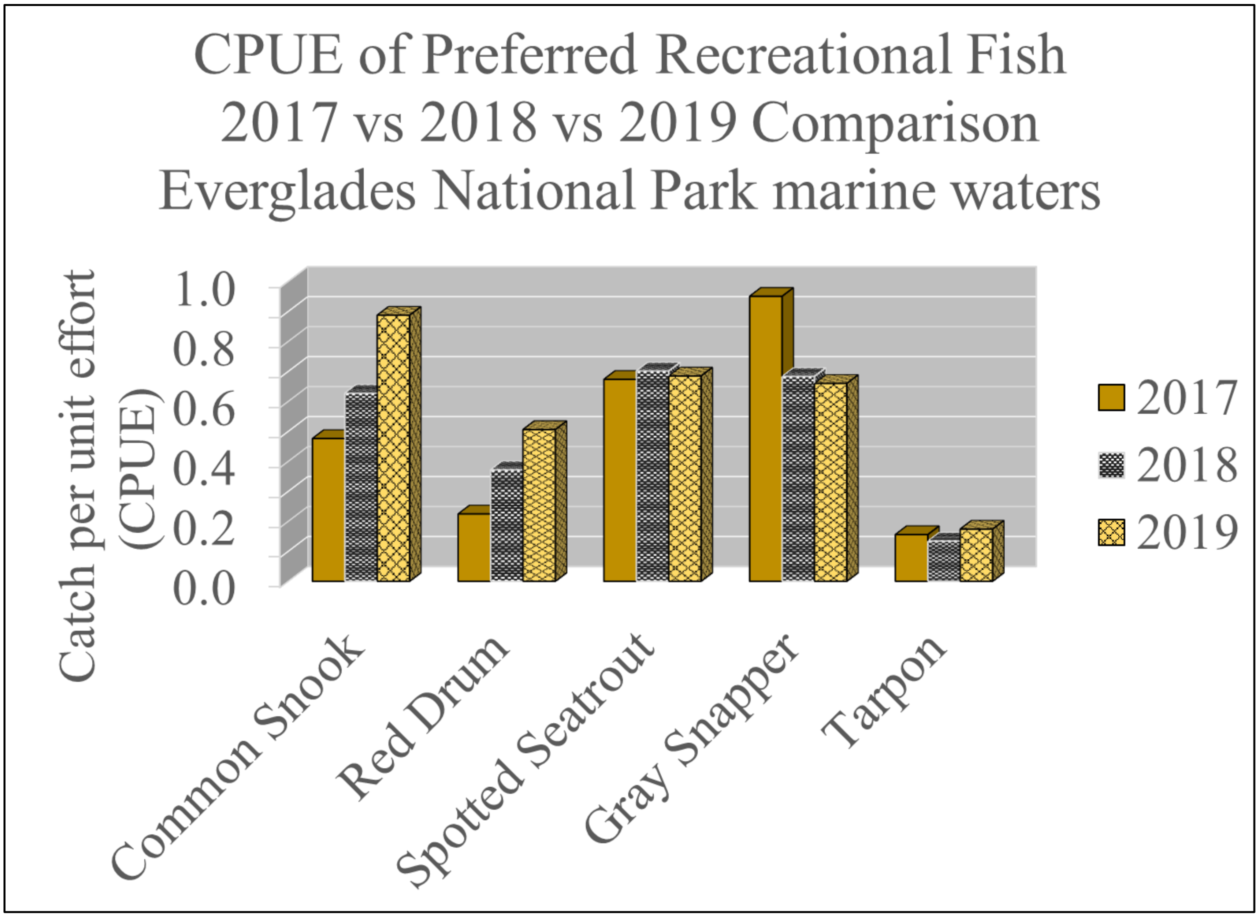

Figure 2. CPUE (fish/hour) of reported hook and line catch of preferred recreational saltwater fish species in Everglades National Park coastal waters for years 2017, 2018 & 2019.

Highlight on Common Snook (*Centropomus undecimalis*)

Snook has shown an extraordinary increase in total catch over the last two years. Numbers of catches increased from 10,707 in 2018 to 17,869 in 2019. Effort, hours fished which resulted in at least one Snook caught, increased 19% from 2018 to 2019. This translated to the highest recreational CPUE (0.89) for *Centropomus undecimalis* for the period of record starting in 1958, and more than double the 2017 CPUE. The vast majority (>70%) of reported Snook catch was from the Gulf Coast region (see Appendices 3, 4 & 5). Anecdotally, the large increase may be associated with a biological response to passage of Hurricane Irma in September 2017, after which observations of large schools of young-of-year Snook and other species were made by researchers and anglers in areas of the coastal zone where they had not been seen in recent years. Most of the Snook catch of the last two years has been described as "under slot" or smaller than legal size limits. The large population increase indicates a favorable recovery from the mortality that occurred during January 2010 when water temperatures in Florida Bay reached 10 degrees Celsius or below for several days (as low as 5°C in some locations), and killed many of this cold-intolerant species in coastal waters of South Florida (Hallac *et al.* 2010, Santos *et al.* 2016, Stevens *et al.* 2016).

Indicator species of ecosystem health

Four species represent indicators of the health of the coastal fishery and comprise most of the preferred catch in Everglades National Park marine and estuarine waters: *Centropomus undecimalis* (Common Snook), *Cynoscion nebulosus* (Spotted Seatrout), *Sciaenops ocellata* (Red Drum), and *Lutjanus griseus* (Gray Snapper). Tarpon are excluded as an indicator due to the infrequency of catch, which may only represent the difficulty in catching this species, as noted by the low CPUE, rather than abundance. The annual changes in reported catch of these four species are used in the assessment of the coastal fishery of the park when reporting the state of conservation of Everglades National Park as a World Heritage site (Table 1). Snook and Red Drum showed increasing trends over the last two years, and positive catch rates indicating healthy or rebounding populations. Snook were regularly reported as abundant, and catch rates approached one fish per angler-hour on trips when Snook were caught. The vast majority of Snook and Red Drum that were caught were reported as smaller "under slot" individuals. Spotted Seatrout catch rates remained stable but persist at more than 20% below the long-term average CPUE from the 1980s and 1990s. Gray Snapper catch rate decreased to a ten-year low in 2019 and remains 10% below the long-term average. Continuing trends of the catch rate for these two species (Spotted Seatrout and Gray Snapper) should be watched with concern.

Table 1. Everglades National Park calendar year 2019 recreational marine fishery indicator species trends in Recreational Catch Per Unit Effort (CPUE). Green solid background with upward arrows indicates conditions are good and improving. Yellow background with horizontal arrows indicates conditions are of moderate concern and stable.

Recreational Catch Per Unit Effort	**Snook (*Centropomus undecimalis*)**	**Red drum (*Sciaenops ocellata*)**	**Spotted seatrout (*Cynoscion nebulosus*)**	**Gray snapper (*Lutjanus griseus*)**
Trend	Snook CPUE exceeded all levels previously recorded over the past six decades in 2019.	Red drum CPUE increased significantly in 2018 and 2019, increasing by 125% over the two years.	Spotted seatrout CPUE remained steady in 2019 but is 23% lower than the average CPUE from the 1980s and 1990s.	Gray snapper CPUE decreased by 28% in 2018 and remained steady in 2019, but is 10% lower than the long-term average.

Additional Catch

The remaining catch of fish included sixty-three species (Appendix 2), of which five species in four families comprised 31% of all reported catch in 2019. Frequently caught fish included Hardhead Catfish, Gafftopsail Catfish (together amounting to 7,351 fish), Ladyfish (5,595 fish), Crevalle Jack (5,014 fish), and Goliath Grouper (964 fish). All other fish caught (2,918 fish) represented less than 5% of the total fish reported and can be grouped practically as those caught commonly (100 or more times per year), incidentally (10 to 99 times per year), and rarely (less than 10 times per year). Commonly caught species included Tarpon, Tripletail, Spanish mackerel, Black Drum, Blacktip and

Bonnethead Sharks, Sheepshead, Lizardfish and Largemouth Bass. Incidental catch comprised twenty-five species and included several species of shark (such as Bull, Lemon, Nurse and Spinner), Smalltooth Sawfish, seatrout (Sand and Silver), jacks and pompano, bluefish, barracuda, kingfish, permit, cobia, other groupers, other snappers, grunts, pinfish, flounder, Mayan Cichlids, and pufferfish. Some of the rare catch included various infrequent species of snapper, jacks, grouper, sharks, stingrays, spadefish, needlefish, toadfish and batfish.

Goliath Grouper (*Epinephelus itajara*)

Also of note in 2019 was the large increase in CPUE of Goliath Grouper in Everglades National Park waters. Reported catch of *Epinephelus itajara* increased from 298 to 964 from 2018 to 2019. In 2017, only 29 Goliath Grouper were reported caught by hook and line from a total of 17 fishing trips in the park. In 2019, Goliath Grouper were caught on 335 sport fishing trips during the year. The 2019 CPUE of 0.22 (Figure 3) was the highest recorded for Goliath Grouper in the park in the last 30 years. Harvest and possession of Goliath Grouper has been prohibited in state and federal waters in the United States southeast fishery since 1990, and the fish caught incidentally must be released immediately alive and unharmed. The Goliath Grouper was listed as critically endangered worldwide on the IUCN Red List in 1994 and is currently listed as Vulnerable (Bertoncini *et al*. 2018, Sadovy de Mitcheson *et al*. 2020). A large proportion (58%) of the Goliath Grouper catch in Everglades National Park normally comes from the northern Gulf Coast area of the park perhaps because the mangroves of the Ten Thousand Island region of Southern Florida have been identified as a nursery habitat for juveniles. Juvenile Goliath Grouper may reside in mangrove coasts for five or six years before moving into deeper waters of the Gulf of Mexico and the reefs of the Atlantic Ocean as adults (Koenig *et al*. 2007, Lara *et al*. 2009). A fairly consistent and much lower percentage of the Goliath Grouper catch comes from Whitewater Bay (8%). The southern area of the Gulf coast usually accounts for 34% of the Goliath Grouper catch with an approximate equal split between the western Florida Bay/Cape Sable region and southern Gulf Coast. Catch in eastern Florida Bay usually consists of single rare individuals but increased to 19 in 2019. An increase in catch has been seen in all areas, but more than 66% of the increase has occurred in the northern Gulf coastal area (from Highland Point to north of Chokoloskee) in the last two years. These data can be considered a good sign for this species of grouper, which was negatively affected by the 2008 and 2010 cold weather events (O'Hop & Munyanderero, 2016).

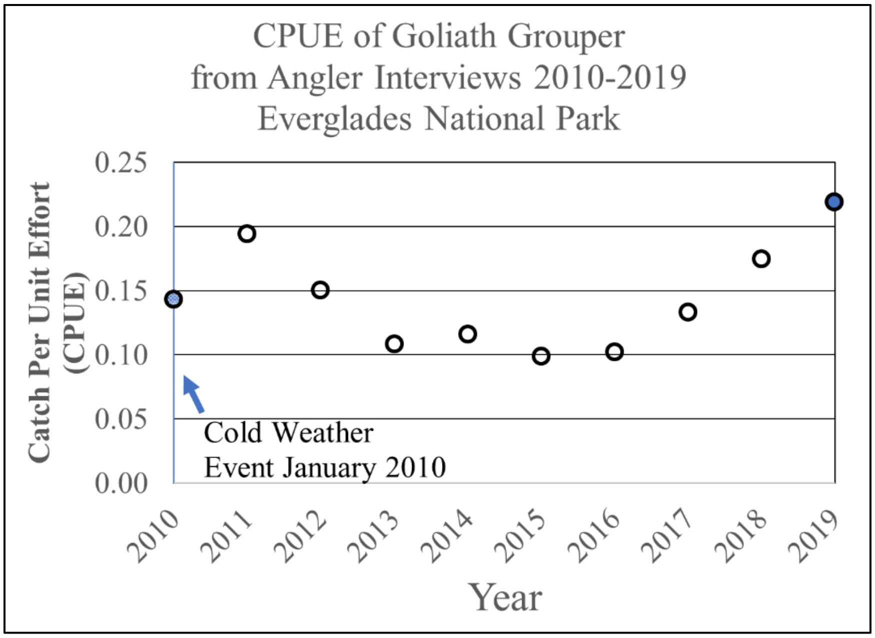

Figure 3. CPUE of Goliath Grouper from 2010 to 2019 based on Creel interviews of anglers at Flamingo and Gulf Coast docks.

Sharks, Rays and Related Species (Cartilaginous fish)

Sharks and rays and relatives (Class Chondrichthyes, the non-bony, cartilaginous fish) are a regular part of the annual catch in Everglades National Park marine waters. Total catch of Chondrichthyes was 32% lower in 2019 than in 2018. Numbers of sharks and related species caught decreased from 641 to 486, with the majority (over 90%) being the following four shark species listed in descending order of CPUE: Blacktip, Bonnethead, Bull and Lemon. Reduction of shark catch occurred primarily through fewer reported Blacktip and Lemon Sharks. Of the sharks reported caught in Everglades National Park, the Dusky, Silky, Lemon, Tiger, Great Hammerhead, and Scalloped Hammerhead Sharks are prohibited take in Florida waters, and must be released immediately without removing the shark from the water or causing injury in removing the hook. If a tagged shark is caught, information from the tag and about the shark should be recorded (tag number, location, species, estimated length, condition and photo if possible) and reported to NOAA National Marine Fisheries Service or the agency on the tag. A small number of Smalltooth Sawfish (*Pristis pectinata*) are reported as incidental catch each year. The Smalltooth Sawfish has been listed as endangered under the U.S. Endangered Species Act since 2003 (Carlson and Simpendorfer 2015), and should not be intentionally caught, harassed or unnecessarily handled. It is currently on the IUCN Red List as "Critically Endangered". If hooked, the sawfish should be released as quickly as possible without injury or removal from the water, and the encounter reported to Everglades National Park with information on the location, size and condition of fish, type of incident, and tag number and photo if possible. NOAA National Marine Fisheries Service monitors the recovery of the Smalltooth Sawfish. Everglades National Park, the Ten Thousand Islands, and Charlotte Harbor marine and estuarine waters have been designated as critical habitat for the species. In 2019, 11 Smalltooth Sawfish were reported as caught in the Everglades National Park.

Other Catch

Fifty-seven identified species represented less than 5% of the total catch in 2019. These species ranged from commonly caught fish reported one hundred or more times such as, Tripletail (*Lobotes surinamensis*; n=440), Spanish Mackerel (Scomberomorus maculatus; n=328), Largemouth Bass (*Micropterus salmoides*; n=306), Black Drum (*Pogonais cromis*; n=148), Sheepshead (Archosargus probatocephalus; n=141), Blacktip Shark (*Carcharhinus limbatus*; n=138) and Bonnethead Shark (*Sphyrna tiburo*; n=110) to single individuals reported only once during the year; for example, batfish, the Silky Shark (*Carcharhinus falciformis*), or Cero (*Scomberomorus regalis*). A total of 33 species were reported caught less than 10 times during the year in 2019 (see Appendices 2 and 3).

Analysis of fish caught, kept and released

The majority of fish caught and reported in Everglades National Park in 2019 were from Gulf Coast waters and the tributary rivers from the western Everglades (37,285 or 60.3%). Florida Bay and Cape Sable represent 28.6% (17,686 fish) of catch and Whitewater Bay another 11.1% (6,872 fish). Of the 61,843 fish caught only 5.85% (3,617) were kept by anglers, resulting in 58,226 fish released after being caught (Appendices 3, 4, 5, and 6).

The three species most often kept in all three areas were Spotted Seatrout, Red Drum and Gray Snapper. Each species has different bag limits, or regulations on numbers allowed to be kept, and size limits. Allowable limits of kept individuals of these species are measured in total length (TL). In 2019, allowable limits of Spotted Seatrout were a maximum of 4 fish >15" to <20" per harvester per day. Regulations for this species changed for 2020 reducing the bag limit to 3 and maximum size <19" (one of the harvestable fish per vessel may exceed 19") for the South Region. Red Drum 2019 regulations allowed one fish harvested/person/day of a size between 18" and 27". Gray Snapper 2019 limits were total length of at least 10" and maximum of 5 fish per harvester.

A total of 21,591 of these three species were caught, and 12% or 2,586 were kept (Table 2). While nearly equal numbers of Spotted Seatrout and Red Drum were caught, over two times the number of Spotted Seatrout were kept. The difference in numbers of seatrout and drum kept are likely due to differences in bag limits, slot size regulations and mean size of the species within the Park. However, Snook is by far the most sought-after fish by anglers in Everglades National Park, according to preferences provided during interviews. Despite the high numbers caught (17,869), less than 0.5% of Snook caught were kept. Both Spotted Seatrout and Red Drum were most frequently reported caught and kept in the Gulf Coast area, and less frequently caught and infrequently kept in Whitewater Bay. The lowest reported total catch and kept numbers were in Whitewater Bay, which may be due to fewer overall anglers or fewer interviews from that launch area. The highest percentage of Grey Snapper kept were from the Florida Bay/Cape Sable area, 22% of those caught. Overall, the highest percentage kept of the three species were Spotted Seatrout at 14.9%, followed by Gray Snapper at 14.6%, then Red Drum at 7.2%.

Table 2. Percentage of most frequently kept fish by area. GC = Gulf Coast, WWB = Whitewater Bay, and FBCS = Florida Bay/Cape Sable.

Area	Common Name	Caught	Kept	Released	% Kept
GC	Red drum	4,916	412	4,504	0.08
FBCG	Red drum	2,711	149	2,562	0.05
WWB	Red drum	401	17	384	0.04
Subtotal	–	8028	578	7,450	7.2%
GC	Spotted seatrout	3,994	663	3,331	0.17
FBCG	Spotted seatrout	3,699	498	3,201	0.13
WWB	Spotted seatrout	865	118	747	0.14
Subtotal	–	8,558	1,279	7,279	14.9%
GC	Gray snapper	2,636	266	2,370	0.10
FBCG	Gray snapper	1,352	304	1,048	0.22
WWB	Gray snapper	1,017	159	858	0.16
Subtotal	–	5,005	729	4,276	14.6%
Total	–	21,591	2,586	19,005	12.0%

A number of species were caught in relatively low numbers but kept with high frequency, including Spanish Mackerel, Sheepshead, Tripletail, Pompano, Black Drum, and Pinfish (in Florida Bay/Cape Sable). Several species were caught in large numbers but kept infrequently, including Snook, Crevalle Jack, Ladyfish, and sea catfish. Nine species make up the majority of kept fish in each of the three areas: Spotted Seatrout, Gray Snapper, Red Drum, Spanish Mackerel, Ladyfish, Crevalle Jack, Tripletail, Sheepshead, and Black Drum.

Fish Length

Special length limits, bag limits, gear restrictions, as well as seasonal and regional restrictions are used to manage and conserve sustainable fish populations in Florida. Length limits may change by region and have a minimum allowable size or a slot size, or a specific range of length required to keep individuals of a named species, for example 28” to 33” for Snook in the Gulf of Mexico and Monroe County. Length of fish, used for state regulations, may be measured as total length or fork length (some species). Most fish length data collected by the Park are to the centermost part of the tailfin in millimeters (mm). Kept fish were measured at the docks where interviews were conducted. In 2019, 1266 fish were measured for length with 85% (1077) being the four preferred species of Snook, Red Drum, Spotted Seatrout and Gray Snapper. Average Snook body length was 707 mm with a minimum of 640 mm and a maximum of 780 mm (Table 3). Red Drum and Spotted Seatrout were on average smaller than Snook due to fishery management size-limit regulations (496 and 434 mm, respectively). Gray Snapper are naturally smaller inshore and averaged 283 mm in length. Analyzing each species across the three fishing areas within the park, revealed small but significant differences in size for Gray Snapper and Spotted Seatrout (Kruskal-Wallis ANOVA on ranks followed by Dunn’s test, α=0.05). Statistical differences in length measures among sites are represented on the graphs with letter groupings of site means (Figure 4). No significant difference in fork length was detected for Snook or Red Drum among GC, WWB, and FBCS areas. Gray Snapper were significantly smaller when taken from GC than either WWB or FBCS. However, Spotted Seatrout were slightly, but significantly, larger when taken from GC than FBCS.

Table 3. Descriptive statistics of body length measurements of four fish species in Everglades National Park.

Species	Sample Size	Mean Length (mm)	Min Length (mm)	Max Length (mm)	Range	Median Length (mm)	25%	75%
Snook	52	707	640	780	140	707	681	725
Red Drum	255	496	428	685	257	482	460	520
Spotted Seatrout	528	434	234	823	589	425	400	453
Gray Snapper	242	283	238	412	174	274	260	295

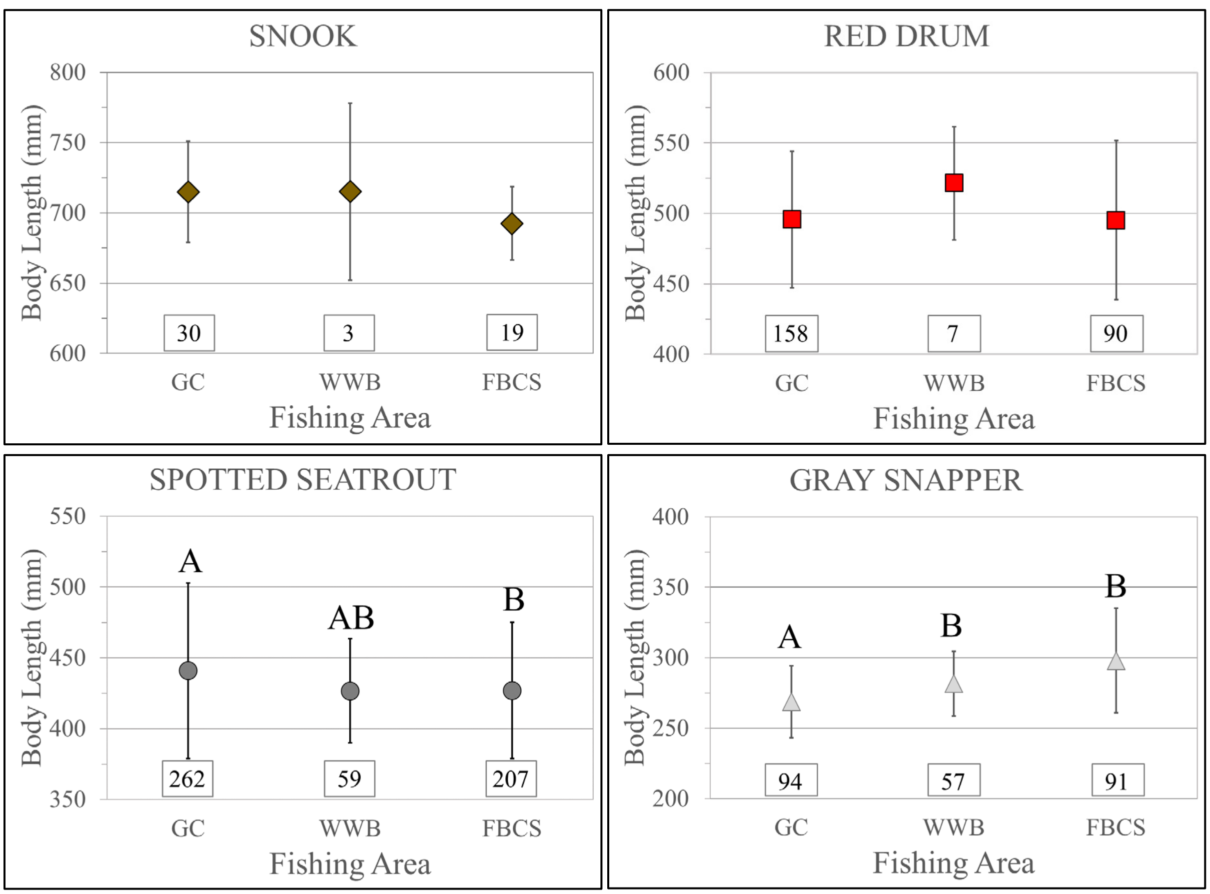

Figure 4. Fork length in millimeters (mm) measured for Snook, Red Drum, Spotted Seatrout, and Gray Snapper. Numbers in boxes are sample sizes. Letters are statistical groupings. Bars are standard deviations. Fishing areas are GC = Gulf Coast, WWB = Whitewater Bay, and FBCS = Florida Bay/Cape Sable.

Invasive Species

At this point in time, the coastal marine system of the Everglades has a single invasive marine fish of high concern that has been observed within the Park. The invasive Indo-Pacific species, *Pterois volitans*, known as the Lionfish, has expanded in range and abundance into the Gulf of Mexico along coastal Florida since 2010. It has been shown to have highly negative impacts to fish populations, consuming, outcompeting and displacing native species. The Lionfish has a formidable defense of venomous spines, and few predators in Florida waters (Figure 5). Its natural habitat is coral reef, but it shows a preference for submerged man-made structures. Rare sightings of Lionfish have been reported in Everglades National Park over the past decade. Over the last two years, opportunistic surveys for presence of Lionfish in Florida Bay have been performed. In 2019, 47 locations were surveyed by snorkeling sites of underwater debris or hardbottom ledges between July and September. A total of 10 juvenile Lionfish, ranging in size from 139 mm to 169 mm, were removed during several cumulative hours of searching. This is fewer fish than during surveys the previous year. In 2018, a total of 35 Lionfish were removed from 12 sites between July and September. The size range was 136 mm to 330 mm, and the average was 237 mm. Individuals of 300 mm length can be considered reproductively mature adults. Lionfish are prolific and can spawn 30,000 eggs every 2–3 days (Gardner 2012) in warm waters once mature. Of forty-five Lionfish captured in two years, more than 75% were hiding under hardbottom ledges, 15% were among submerged man-made debris. These results indicate that Lionfish are not yet widespread in the shallow water habitat of Everglades National Park in great numbers, while much of coastal Florida has been overwhelmed by this invasive species. If a Lionfish is observed while boating or fishing, report the location to Park marine fisheries staff. If caught, Lionfish should not be released, but care must be taken to avoid the venomous spines.

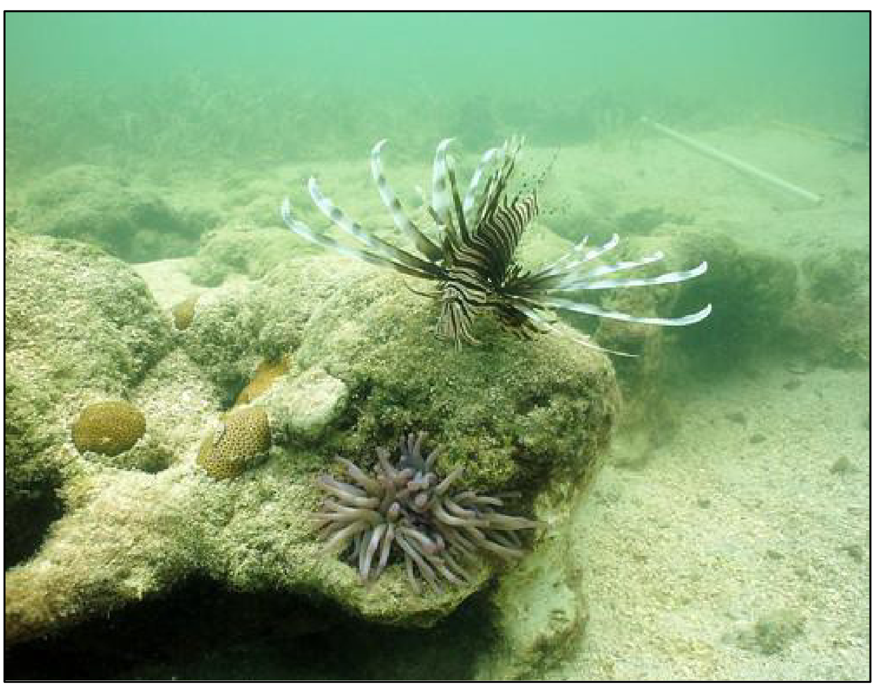

Figure 5. Lionfish, *Pterois volitans*. Photo by Robert Ellis, FWRI, with permission.

Visitor use data

Peak angler activity as measured by numbers of fishing trips interviewed per month was seen in February, March and November of 2019. The number of fishing trip interviews used to collect the fishing data in this report totaled 2,461 (Figure 6) including both sites at which Creel interviews were performed (Flamingo and Everglades City). Most of January interview data was missing in 2019 due to the federal government closure, which lasted from December 22, 2018 through January 25, 2019. A limited number of interviews were conducted on one day in January at the Flamingo site. The total number of fishing trips interviewed in 2019 was 4% less than in 2018, and the difference may represent the number of interviews missed in January. The average number of anglers per trip was 2.17, which has dropped steadily in the past five years. The average number of anglers per trip in the previous five-year period (2010–2014) was stable at 2.33. The number of reported hours fished decreased by 4% from 2108 to 2019 (14,445 to 13,900). However, the success rate for fishing trips (the percentage of reported trips resulting in at least one fish caught during a trip) increased from 93.3% to 96.8% from 2018 to 2019. An approximate equal number of fishing trips were recorded for each interview site; 53% Gulf Coast and 47% Flamingo. During interviews, the residency of anglers was categorized as either from the local area, within the state of Florida but outside the local area, and out of the state of Florida.

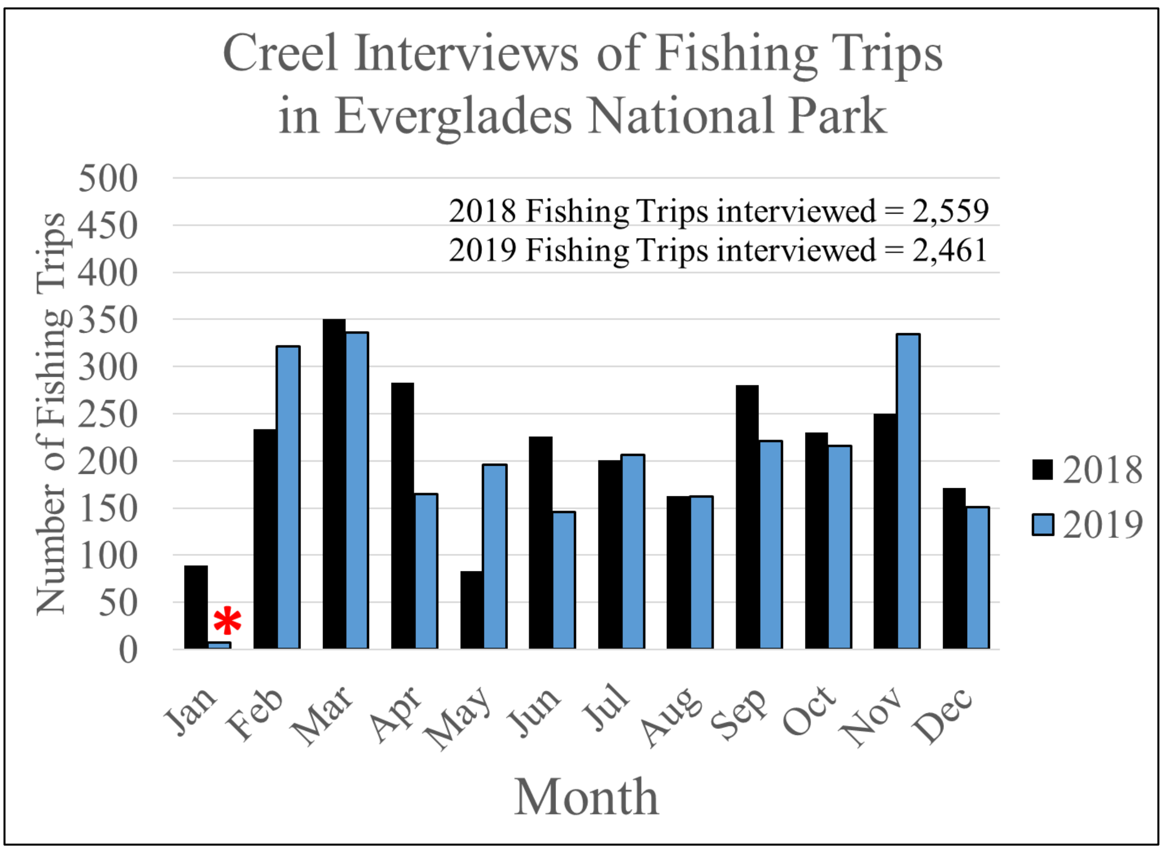

Figure 6. Total fishing trips interviewed per month in Creel surveys at Flamingo and Everglades City docks for 2018 and 2019. *January 2019 included interviews from only one day due to federal government closure in January.

Results from 2019 (Figure 7) show a shift in the predominant residency of interviewed anglers fishing within the park, with the number of Florida resident anglers from outside the local area exceeding those from the local area. The percentage of local area anglers interviewed declined to 46% of the total conducted in 2019, which is significantly lower than the long-term average of greater than 50%. The concurrent rise in Florida residents from outside the local area angling in the park reached 54% in 2019. The percentage of anglers from outside the state changed little from 2018 to 2019.

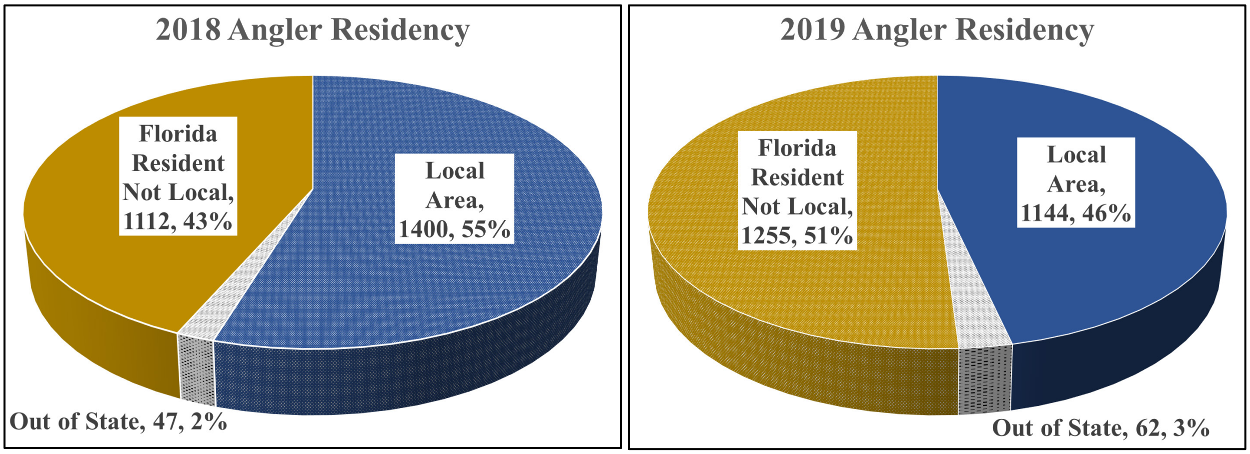

Figure 7. Visiting angler residency by percentage of total trips interviewed in Everglades National Park. Categories included Local Area, Out of State and Florida resident-not local.

Conclusion

Reported fish catch increased substantially in Everglades National Park from 2018 to 2019. CPUE of Snook reached a 60-year record high, primarily due to an abundance of smaller-than-slot sized fish. Red Drum also showed a significant increase in catch and CPUE (with similar reports on size of released fish). An increase of Goliath Grouper catch in the Gulf of Mexico waters of the park indicated a possible increase in the population of this protected species. Trends in CPUE of Spotted Seatrout and Gray Snapper were neutral from 2018 to 2019, although some concern remains since the catch is lower than the long-term average of each species. Gray Snapper CPUE is down significantly from a 35-year high mark in 2017. Relatively few fish caught are actually kept by anglers, either due to regulatory limits or by choice. Only 3,617 of the total 61,843 fish reported caught, less than 6%, were kept by anglers.

The increases in catch rate of the preferred fish accompanied an increase in fishing success rate, which rose from 93.3% to 96.9% from 2018 to 2019. Data concerning anglers from the 2019 season indicated a continued long-term trend representing a decline in the number of anglers per trip. A significant shift in angler residency was recorded in 2019. Fewer local anglers compared to Florida residents from outside the local area were interviewed for the first time in the period of record.

The status of the invasive species *Pterois volitans* in park waters has yet to be fully assessed, but preliminary surveys suggest that relatively few Lionfish have become established in Florida Bay within the park. Reported incidental catch of the endangered species, *Pristis pectinata*, totaled 11 fish, primarily in Gulf Coast waters.

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Appendix 1. Reported catch of interviewed anglers by fish family of saltwater species in Everglades National Park in 2019.

Family	Common Name	Catch
Centropomidae	Common snook	17869
Sciaenidae	Seatrout, Red & Black Drum	16900
Ariidae	Catfish: Hardheads & Gafftopsail	7351
Elopidae	Tarpon & Ladyfish	6133
Carangidae	Jacks, Pompano, Runners, Permit	5191
Lutjanidae	Snapper	5040
Serranidae	Grouper	988
Lobotidae	Tripletail	440
Scombridae	Mackerel	330
Carcharinidae	Requiem Sharks	329
Centrarchidae	Largemouth Bass	306
Sparidae	Sheepshead & Pinfish	213
Synodontidae	Lizardfish	138
Sphyrnidae	Bonnethead & Hammerhead Sharks	113
Pomatomidae	Blues	92
Mugilidae	Mullet	86
Tetraodontidae	Pufferfish	69
Cichlidae	Mayan Cichlid	49
Sphyraenidae	Barracuda	48
Bothidae	Left-eye Flounder	46
Dasyatidae	Stingrays	21
Rachycentridae	Cobia	17
Pomadasyidae	Grunts	14
Ginglymostomatidae	Nurse Sharks	12
Pristidae	Smalltooth Sawfish	11
Lepisosteidae	Gars	7
Belonidae	Needlefish, Houndfish	6
Echeneidae	Remoras	6
Batrachoididae	Toadfish	5
Portunidae	Blue crab	5
Ephippidae	Spadefish	2
Gerreidae	Mojarra	2
Ogcocephalidae	Batfish	1

Appendix 2. Catch of saltwater species in Everglades National Park in 2019.

Family	Genus	Species	Common Name	Total Caught
Centropomidae	Centropomus	undecimalis	Snook	17869
Sciaenidae	Cynoscion	nebulosus	Spotted seatrout	8558
Sciaenidae	Sciaenops	ocellata	Red drum	8028
Elopidae	Elops	saurus	Ladyfish	5595
Carangidae	Caranx	hippos	Crevalle jack	5014
Lutjanidae	Lutjanus	griseus	Gray (Mangrove) snapper	5005
Ariidae	unknown	unknown	Catfish	4528
Ariidae	Arius	felis	Hardheads catfish	1568
Ariidae	Bagre	marinus	Gafftopsail catfish	1255
Serranidae	Epinephelus	itajara	Goliath grouper	964
Elopidae	Megalops	atlantica	Tarpon	538
Lobotidae	Lobotes	surinamensis	Atlantic tripletail	440
Scombridae	Scomberomorus	maculatus	Spanish mackerel	328
Centrarchidae	Micropterus	salmoides	Largemouth bass	306
Sciaenidae	Pogonias	cromis	Black drum	148
Sparidae	Archosargus	probatocephalus	Sheepshead	141
Carcharhinidae	Carcharhinus	limbatus	Blacktip shark	138
Synodontidae	unknown	unknown	Lizardfish	138
Sphyrnidae	Sphyrna	tiburo	Bonnethead shark	110
Carangidae	Caranx	crysos	Blue runner	94
Pomatomidae	Pomatomus	saltatrix	Bluefish	92
Mugilidae	unknown	unknown	Mullet	86
Sciaenidae	Cynoscion	arenarius	Sand seatrout	82
Carcharhinidae	Carcharhinus	leucas	Bull shark	76
Sparidae	Lagodon	rhomboides	Pinfish	72
Tetraodontidae	unknown	unknown	Pufferfish	69
Carcharhinidae	Negaprion	brevirostris	Lemon shark	54
Carangidae	Trachinotus	carolinus	Florida pompano	52
Cichlidae	Cichlasoma	urophthalmus	Mayan cichlid	49
Bothidae	unknown	unknown	Left-eye flounders	46
Sphyraenidae	Sphyraena	barracuda	Great barracuda	46
Carcharhinidae	unknown	unknown	Requiem Sharks	43
Sciaenidae	Cynoscion	nothus	Silver seatrout	32
Sciaenidae	Menticirrhus	littoralis	Gulf kingfish	30
Carangidae	Trachinotus	falcatus	Permit	22

Family	Genus	Species	Common Name	Total Caught
Dasyatidae	unknown	unknown	Whiptail stingrays	18
Rachycentridae	Rachycentron	canadum	Cobia	17
Serranidae	Mycteroperca	microlepis	Gag grouper	17
Lutjanidae	Lutjanus	synagris	Lane snapper	14
Pomadasyidae	unknown	unknown	Grunts	14
Sciaenidae	unknown	unknown	Drums or croakers	14
Ginglymostomatidae	Ginglymostoma	cirratum	Nurse shark	12
Pristidae	Pristis	pectinata	Smalltooth sawfish	11
Carcharhinidae	Carcharhinus	maculipinnis	Spinner shark	10

Appendix 3. Catch of rare saltwater species in Everglades National Park in 2019.

Family	Genus	Species	Common Name	Total Caught
Carcharhinidae	Rhizoprionodon	terraenovae	Atlantic sharpnose shark	7
Lutjanidae	Lutjanus	apodus	Schoolmaster snapper	7
Carangidae	Selene	vomer	Lookdown fish	6
Echeneidae	unknown	unknown	Remoras	6
Batrachoididae	unknown	unknown	Toad Fish	5
Portunidae	Callinectes	sapidus	Blue crab	5
Lutjanidae	unknown	unknown	Snappers	5
Lepisosteidae	unknown	unknown	Gars	4
Sciaenidae	Cynoscion	regalis	Weakfish	4
Serranidae	Mycteroperca	bonaci	Black grouper	4
Belonidae	unknown	unknown	Needlefishes	3
Belonidae	Tylosurus	crocodilus	Hound fish	3
Dasyatidae	Dasyatis	americana	Southern stingray	3
Lepisosteidae	Lepisosteus	platyrhincus	Florida gar	3
Lutjanidae	Lutjanus	analis	Mutton snapper	3
Lutjanidae	Lutjanus	jocu	Dog snapper	3
Sphyrnidae	unknown	unknown	Hammerhead sharks	3
Carangidae	Oligoplites	saurus	Leather jack, Leatherjacket	2
Ephippidae	Chaetodipterus	faber	Atlantic spadefish	2
Gerreidae	Eucinostomus	gula	Silvery mojarra	2
Lutjanidae	Lutjanus	campechanus	Red snapper	2
Sciaenidae	Micropogonias	undulatus	Atlantic croaker	2
Serranidae	unknown	unknown	Groupers	2
Sphyraenidae	unknown	unknown	Barracudas	2
Carangidae	unknown	unknown	Jacks, pompano, runners	1
Carangidae	Caranx	latus	Horse eye jack	1
Carcharhinidae	Carcharhinus	falciformis	Silky shark	1
Lutjanidae	Lutjanus	cyanopterus	Cubera snapper	1
Ogcocephalidae	unknown	unknown	Batfish	1
Sciaenidae	Bairdiella	chrysura	Silver perch	1
Scombridae	Euthynnus	alletteratus	Little tunny	1
Scombridae	Scomberomorus	regalis	Cero, Cero mackerel	1
Serranidae	Epinephelus	morio	Red grouper	1

Appendix 4. Gulf Coast species caught, kept, released in 2019 (descending order by catch).

Family	Genus	Species	Common	Caught	Kept	Released	% Kept
Centropomidae	Centropomus	undecimalis	Snook	12608	52	12556	0.4%
Sciaenidae	Sciaenops	ocellata	Red drum	4916	412	4504	8.4%
Sciaenidae	Cynoscion	nebulosus	Spotted seatrout	3994	663	3331	16.6%
Elopidae	Elops	saurus	Ladyfish	3751	87	3664	2.3%
Lutjanidae	Lutjanus	griseus	Gray snapper	2636	266	2370	10.1%
Carangidae	Caranx	hippos	Crevalle jack	2555	26	2529	1.0%
Ariidae	unknown	unknown	sea catfishes	1584	0	1584	0.0%
Ariidae	Arius	felis	Hardhead catfish	1355	0	1355	0.0%
Ariidae	Bagre	marinus	Gafftopsail catfish	972	19	953	2.0%
Serranidae	Epinephelus	itajara	Goliath grouper	834	2	832	0.2%
Centrarchidae	Micropterus	salmoides	Largemouth bass	289	0	289	0.0%
Lobotidae	Lobotes	surinamensis	Tripletail	205	47	158	22.9%
Elopidae	Megalops	atlantica	Tarpon	188	0	188	0.0%
Scombridae	Scomberomorus	maculatus	Spanish mackerel	185	104	81	56.2%
Sparidae	Archosargus	probatocephalus	Sheepshead	113	46	67	40.7%
Sciaenidae	Pogonias	cromis	Black drum	99	34	65	34.3%
Pomatomidae	Pomatomus	saltatrix	Bluefish	85	8	77	9.4%
Mugilidae	unknown	unknown	mullets	80	70	10	87.5%
Sciaenidae	Cynoscion	arenarius	Sand seatrout	80	21	59	26.3%
Carcharhinidae	Carcharhinus	limbatus	Blacktip shark	79	0	79	0.0%
Carangidae	Caranx	crysos	Blue runner	73	13	60	17.8%
Synodontidae	unknown	unknown	lizardfishes	70	0	70	0.0%
Sphyrnidae	Sphyrna	tiburo	Bonnethead	68	0	68	0.0%
Carcharhinidae	Carcharhinus	leucas	Bull shark	55	1	54	1.8%
Cichlidae	Cichlasoma	urophthalmus	Mayan cichlid	49	0	49	0.0%
Bothidae	unknown	unknown	lefteye flounders	46	0	3	0.0%
Carangidae	Trachinotus	carolinus	Florida pompano	43	31	12	72.1%
Sciaenidae	Cynoscion	nothus	Silver seatrout	32	6	26	18.8%
Tetraodontidae	unknown	unknown	puffers	32	0	32	0.0%
Carcharhinidae	unknown	unknown	requiem sharks	23	0	23	0.0%
Carangidae	Trachinotus	falcatus	Permit	17	5	12	29.4%
Sciaenidae	Menticirrhus	littoralis	Gulf kingfish	17	9	8	52.9%
Serranidae	Mycteroperca	microlepis	Gag grouper	15	1	14	6.7%
Dasyatidae	unknown	unknown	stingrays	14	0	14	0.0%
Carcharhinidae	Negaprion	brevirostris	Lemon shark	13	1	12	7.7%
Sciaenidae	unknown	unknown	drums	13	0	13	0.0%
Ginglymostomatidae	Ginglymostoma	cirratum	Nurse shark	10	0	10	0.0%
Lutjanidae	Lutjanus	synagris	Lane snapper	9	2	7	22.2%

Family	**Genus**	**Species**	**Common**	**Caught**	**Kept**	**Released**	**% Kept**
Pristidae	Pristis	pectinata	Smalltooth sawfish	9	0	9	0.0%
Pomadasyidae	unknown	unknown	grunts	8	3	5	37.5%
Rachycentridae	Rachycentron	canadum	Cobia	8	3	5	37.5%
Carcharhinidae	Rhizoprionodon	terraenovae	Atlantic sharpnose shark	6	0	6	0.0%
Sphyraenidae	Sphyraena	barracuda	Great barracuda	6	0	6	0.0%
Batrachoididae	unknown	unknown	toadfishes	5	0	5	0.0%
Serranidae	Mycteroperca	bonaci	Black grouper	4	0	4	0.0%
Carcharhinidae	Carcharhinus	maculipinnis	Spinner shark	3	0	3	0.0%
Lepisosteidae	Lepisosteus	platyrhincus	Florida gar	3	0	3	0.0%
Lutjanidae	unknown	unknown	snappers	3	0	3	0.0%
Sphyrnidae	unknown	unknown	hammerhead sharks	3	0	3	0.0%
Gerreidae	Eucinostomus	gula	Silver jenny	2	2	0	100.0%
Lepisosteidae	unknown	unknown	gars	2	0	2	0.0%
Lutjanidae	Lutjanus	jocu	Dog snapper	2	0	2	0.0%
Serranidae	unknown	unknown	groupers	2	0	2	0.0%
Sphyraenidae	unknown	unknown	barracudas	2	0	2	0.0%
Belonidae	Tylosurus	crocodilus	Houndfish	1	0	1	0.0%
Belonidae	unknown	unknown	needlefishes	1	0	1	0.0%
Carangidae	Oligoplites	saurus	Leatherjacket	1	0	1	0.0%
Carangidae	Selene	vomer	Lookdown	1	0	1	0.0%
Echeneidae	unknown	unknown	remoras	1	0	1	0.0%
Ogcocephalidae	unknown	unknown	batfishes	1	0	1	0.0%
Portunidae	Callinectes	sapidus	Blue crab	1	0	1	0.0%
Scombridae	Euthynnus	alletteratus	Little tunny	1	0	1	0.0%
Scombridae	Scomberomorus	regalis	Cero	1	0	1	0.0%
Serranidae	Epinephelus	morio	Red grouper	1	0	1	0.0%
Total	–	–	–	37285	1934	35308	5.2%

Appendix 5. Florida Bay/Cape Sable species caught, kept, and released in 2019 (descending order by catch).

Family	Genus	Species	Common	Caught	Kept	Released	% Kept
Sciaenidae	Cynoscion	nebulosus	Spotted seatrout	3699	498	3201	13.5%
Centropomidae	Centropomus	undecimalis	Snook	2741	25	2716	0.9%
Sciaenidae	Sciaenops	ocellata	Red drum	2711	149	2562	5.5%
Ariidae	unknown	unknown	sea catfishes	2519	19	2500	0.8%
Carangidae	Caranx	hippos	Crevalle jack	2027	58	1969	2.9%
Lutjanidae	Lutjanus	griseus	Gray snapper	1352	304	1048	22.5%
Elopidae	Elops	saurus	Ladyfish	1159	57	1102	4.9%
Ariidae	Bagre	marinus	Gafftopsail catfish	229	21	208	9.2%
Elopidae	Megalops	atlantica	Tarpon	212	0	212	0.0%
Lobotidae	Lobotes	surinamensis	Tripletail	207	38	169	18.4%
Ariidae	Arius	felis	Hardhead catfish	155	0	155	0.0%
Scombridae	Scomberomorus	maculatus	Spanish mackerel	126	76	50	60.3%
Sparidae	Lagodon	rhomboides	Pinfish	72	32	40	44.4%
Serranidae	Epinephelus	itajara	Goliath grouper	68	0	68	0.0%
Carcharhinidae	Carcharhinus	limbatus	Blacktip shark	52	0	52	0.0%
Sciaenidae	Pogonias	cromis	Black drum	44	15	29	34.1%
Sphyrnidae	Sphyrna	tiburo	Bonnethead	42	0	42	0.0%
Carcharhinidae	Negaprion	brevirostris	Lemon shark	40	0	40	0.0%
Sphyraenidae	Sphyraena	barracuda	Great barracuda	31	0	31	0.0%
Sparidae	Archosargus	probatocephalus	Sheepshead	23	12	11	52.2%
Carangidae	Caranx	crysos	Blue runner	21	1	20	4.8%
Carcharhinidae	unknown	unknown	requiem sharks	18	0	18	0.0%
Sciaenidae	Menticirrhus	littoralis	Gulf kingfish	13	2	11	15.4%
Carcharhinidae	Carcharhinus	leucas	Bull shark	11	0	11	0.0%
Carangidae	Trachinotus	carolinus	Florida pompano	9	3	6	33.3%
Synodontidae	unknown	unknown	lizardfishes	9	0	9	0.0%
Rachycentridae	Rachycentron	canadum	Cobia	8	5	3	62.5%
Carcharhinidae	Carcharhinus	maculipinnis	Spinner shark	7	0	7	0.0%
Pomatomidae	Pomatomus	saltatrix	Bluefish	7	3	4	42.9%
Lutjanidae	Lutjanus	apodus	Schoolmaster	6	3	3	50.0%
Mugilidae	unknown	unknown	mullets	6	5	1	83.3%
Pomadasyidae	unknown	unknown	grunts	6	0	6	0.0%
Carangidae	Trachinotus	falcatus	Permit	5	1	4	20.0%
Lutjanidae	Lutjanus	synagris	Lane snapper	5	2	3	40.0%
Carangidae	Selene	vomer	Lookdown	4	0	4	0.0%
Echeneidae	unknown	unknown	remoras	4	0	4	0.0%
Portunidae	Callinectes	sapidus	Blue crab	4	0	4	0.0%
Dasyatidae	Dasyatis	americana	Southern stingray	3	0	3	0.0%

Family	Genus	Species	Common	Caught	Kept	Released	% Kept
Dasyatidae	unknown	unknown	stingrays	3	0	3	0.0%
Lutjanidae	Lutjanus	analis	Mutton snapper	3	0	3	0.0%
Belonidae	Tylosurus	crocodilus	Houndfish	2	0	2	0.0%
Lutjanidae	Lutjanus	campechanus	Red snapper	2	1	1	50.0%
Ginglymostomatidae	Ginglymostoma	cirratum	Nurse shark	2	0	2	0.0%
Pristidae	Pristis	pectinata	Smalltooth sawfish	2	0	2	0.0%
Sciaenidae	Cynoscion	arenarius	Sand seatrout	2	1	1	50.0%
Sciaenidae	Micropogonias	undulatus	Atlantic croaker	2	0	2	0.0%
Belonidae	unknown	unknown	needlefishes	1	0	1	0.0%
Carangidae	Caranx	latus	Horse-eye jack	1	0	1	0.0%
Carangidae	Oligoplites	saurus	Leatherjacket	1	0	1	0.0%
Carangidae	unknown	unknown	jacks and pompanos	1	0	1	0.0%
Carcharhinidae	Carcharhinus	falciformis	Silky shark	1	0	1	0.0%
Carcharhinidae	Rhizoprionodon	terraenovae	Atlantic sharpnose shark	1	0	1	0.0%
Lepisosteidae	unknown	unknown	gars	1	0	1	0.0%
Lutjanidae	Lutjanus	cyanopterus	Cubera snapper	1	0	1	0.0%
Lutjanidae	Lutjanus	jocu	Dog snapper	1	1	0	100.0%
Lutjanidae	unknown	unknown	snappers	1	0	1	0.0%
Sciaenidae	Bairdiella	chrysura	Silver perch	1	0	1	0.0%
Sciaenidae	Cynoscion	regalis	Weakfish	1	0	1	0.0%
Serranidae	Mycteroperca	microlepis	Gag grouper	1	0	1	0.0%
Total	–	–	–	17686	1332	16354	7.5%

Appendix 6. Whitewater Bay species caught, kept, and released in 2019 (descending order by catch).

Family	Genus	Species	Common	Caught	Kept	Released	% Kept
Centropomidae	Centropomus	undecimalis	Snook	2520	5	2515	0.2%
Lutjanidae	Lutjanus	griseus	Gray snapper	1017	159	858	15.6%
Sciaenidae	Cynoscion	nebulosus	Spotted seatrout	865	118	747	13.6%
Elopidae	Elops	saurus	Ladyfish	685	7	678	1.0%
Carangidae	Caranx	hippos	Crevalle jack	432	0	432	0.0%
Ariidae	unknown	unknown	sea catfishes	425	0	425	0.0%
Sciaenidae	Sciaenops	ocellata	Red drum	401	17	384	4.2%
Elopidae	Megalops	atlantica	Tarpon	138	0	138	0.0%
Serranidae	Epinephelus	itajara	Goliath grouper	62	0	62	0.0%
Synodontidae	unknown	unknown	lizardfishes	59	0	59	0.0%
Ariidae	Arius	felis	Hardhead catfish	58	0	58	0.0%
Ariidae	Bagre	marinus	Gafftopsail catfish	54	0	54	0.0%
Tetraodontidae	unknown	unknown	puffers	37	0	37	0.0%
Lobotidae	Lobotes	surinamensis	Tripletail	28	6	22	21.4%
Scombridae	Scomberomorus	maculatus	Spanish mackerel	18	6	12	33.3%
Centrarchidae	Micropterus	salmoides	Largemouth bass	17	0	17	0.0%
Carcharhinidae	Carcharhinus	leucas	Bull shark	10	1	9	10.0%
Sphyraenidae	Sphyraena	barracuda	Great barracuda	9	0	9	0.0%
Carcharhinidae	Carcharhinus	limbatus	Blacktip shark	7	0	7	0.0%
Sciaenidae	Pogonias	cromis	Black drum	6	0	6	0.0%
Sparidae	Archosargus	probatocephalus	Sheepshead	5	2	3	40.0%
Sciaenidae	Cynoscion	regalis	Weakfish	3	0	3	0.0%
Carcharhinidae	unknown	unknown	requiem sharks	2	0	2	0.0%
Ephippidae	Chaetodipterus	faber	Atlantic spadefish	2	0	2	0.0%
Belonidae	unknown	unknown	needlefishes	1	0	1	0.0%
Carangidae	Selene	vomer	Lookdown	1	0	1	0.0%
Carcharhinidae	Negaprion	brevirostris	Lemon shark	1	0	1	0.0%
Dasyatidae	unknown	unknown	stingrays	1	0	1	0.0%
Echeneidae	unknown	unknown	remoras	1	0	1	0.0%
Lepisosteidae	unknown	unknown	gars	1	0	1	0.0%
Lutjanidae	Lutjanus	apodus	Schoolmaster	1	1	0	100.0%
Lutjanidae	Lutjanus	cyanopterus	Cubera snapper	1	1	0	100.0%
Lutjanidae	unknown	unknown	snappers	1	1	0	100.0%
Rachycentridae	Rachycentron	canadum	Cobia	1	1	0	100.0%
Sciaenidae	unknown	unknown	drums	1	0	1	0.0%
Serranidae	Mycteroperca	microlepis	Gag grouper	1	0	1	0.0%
Total	–	–	–	6872	325	6547	4.7%

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